

Spatial Point Pattern Analysis



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Spatial Point Pattern Analysis

Introduction:

In this lab, you will learn how to create a change detection map using species observation data collected from iNaturalist, a citizen science platform. You will analyze how spatial patterns of observations change over time by applying point pattern analysis techniques.

Objectives:

1. Gain familiarity with citizen science platform (iNaturalist)
2. Download data from iNaturalist and visualize them in ArcGIS Pro
3. Map species concentrations using Thiessen polygons and Kernel Density Estimation
4. Detect and analyze changes in species distribution over time

A short introduction of iNaturalist:

- iNaturalist is an online platform for recording, sharing, and identifying nature observations.
- Founded in 2008 as a UC Berkeley Master's project, it now hosts a large global user community.
- Participants upload photos of plants, animals, insects, and other organisms.
- Computer vision models suggest identifications, which are refined through community input and expert review.
- All observations contribute to publicly available biodiversity data, supporting science and conservation.

1. Data:

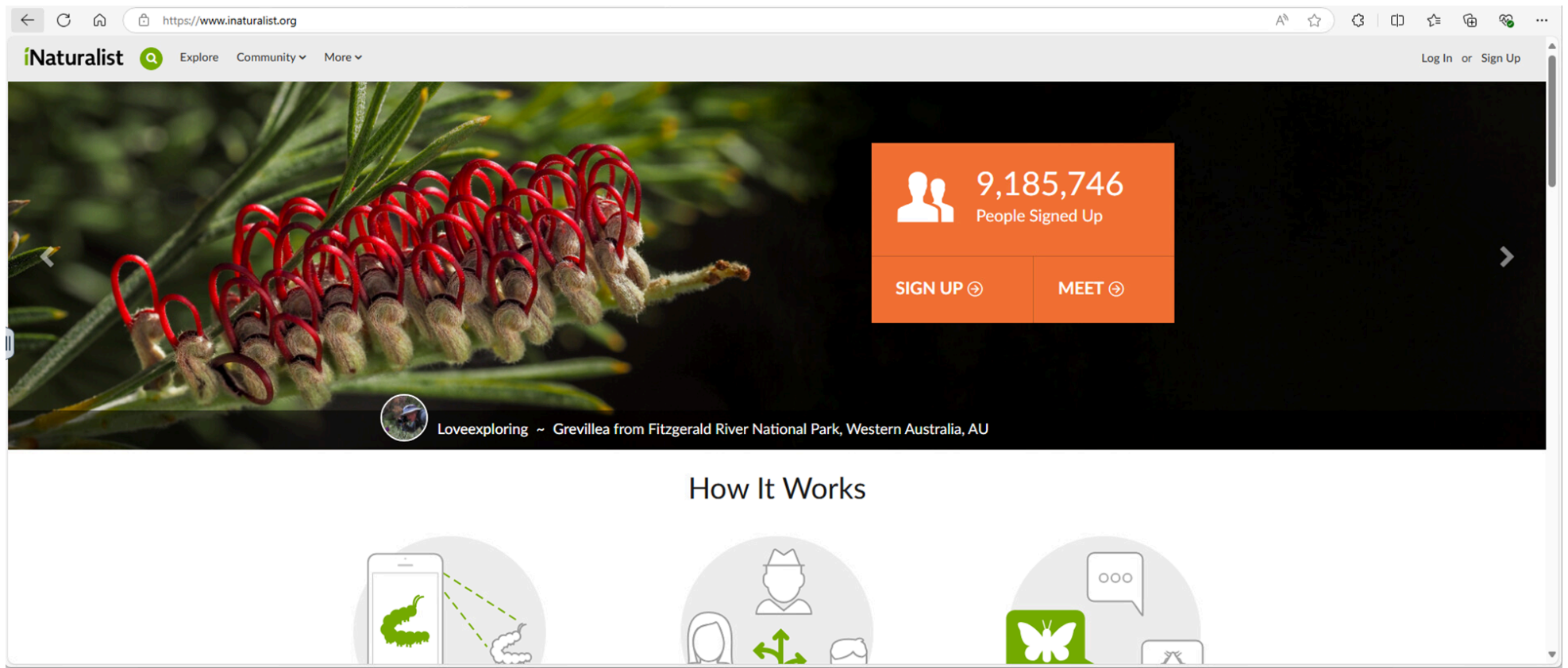
Answer Sheet : [Answer Sheet.docx \(https://ufl.instructure.com/courses/541369/files/99937768?wrap=1\)](https://ufl.instructure.com/courses/541369/files/99937768?wrap=1)

[observations-2023.csv \(https://ufl.instructure.com/courses/541369/files/99942569?wrap=1\)](https://ufl.instructure.com/courses/541369/files/99942569?wrap=1) ↓
(https://ufl.instructure.com/courses/541369/files/99942569/download?download_frd=1)

[observations-2024.csv \(https://ufl.instructure.com/courses/541369/files/99942604?wrap=1\)](https://ufl.instructure.com/courses/541369/files/99942604?wrap=1) ↓
(https://ufl.instructure.com/courses/541369/files/99942604/download?download_frd=1)

You will work with two CSV files of rattlesnake observations from 2023 and 2024. These are provided for convenience, as downloading data from iNaturalist can sometimes take 1–2 hours depending on size of data. If you use the provided files, skip ahead to part **2. ArcGIS Pro**. If you prefer to use your own species data, follow the instructions below to download it from iNaturalist.

1. Sign up for an account on the iNaturalist website: <https://www.inaturalist.org/> ↗ (<https://www.inaturalist.org/>)



2. Choose a species you're interested in and enter its name in the iNaturalist search bar—for example, "rattlesnakes." Then, click "Filters" to refine your search.

iNaturalist Explore Community More Log In or Sign Up

Observations

Rattlesnakes
Genus Crotalus [View](#)

Massasaugas (Pygmy rattlesnakes)
Genus Sistrurus [View](#)

272,770 OBSERVATIONS **4** IERS **3,860,322** OBSERVERS

Red-striped Needleworm Moth
(*Epinotia radicans*)
malson Ville-Marie... Yesterday
1 1h

White Underwing
(*Catocala relictal*)
malson Ville-Marie... Yesterday
1 1h

Black-letter Darts
(*Complex Xestia c-nigrum*)
malson Ville-Marie... Sep 8, 2025
1 1h

Venerable Dart
(*Agrotis venerabilis*)
malson Ville-Marie... Sep 7, 2025
1 1h

Venerable Dart

Keyboard shortcuts | Map data ©2025 | 1000 km | Terms

3. Set your search criteria to display all rattlesnake observations in the United States during 2023.

- Specify the date range: **2023-01-01 to 2023-12-31**
- Set the Place to: **United States**
- Select **Research Grade** (i.e. observations that meet a minimum threshold of quality)
- Click **Download**

Observations

United States

Map

Grid

List

+

-

📍

📏

Map Legend

Rattlesnakes

Location

Go

Filters

Show

☐ Wild
☐ Captive
☐ Verifiable
☒ Research Grade
☐ Needs ID
☐ Threatened

Categories

🐞

🐜

🐛

🐌

🐚

🐜

🐜

🐜

🐜

🐜

☐ Introduced
☐ Popular
☐ Has Sounds
☐ Has Photos

Rank

High

Low

Sort By

Date Added

Desc

Date Observed

☐ Any
☐ Exact Date
☒ Range

☐ Months

Select Options

Description / Tags

blue, butterfly, etc.

More Filters

Person

👤

Username or User ID

Project

📁

Name or URL slug, e.g. my-project

Place

🌐

United States

Photo Licensing

All

Reviewed

☒ Any
☐ Yes
☐ No

Date Added

☒ Any
☐ Exact Date
☐ Range

Update Search

Reset Search Filters

Identify

Atom

Download

4. On the “Export Observations” page, go to Part 3 under “Geo” and select all columns to ensure you include complete geographic data.

Geo (All | None)

Town, county, state, country fields based on coordinates. Analogous administrative levels will be used for observations where these place types do not apply. The private_* fields will only be included when using the user filter and you are the user, or when using the project filter and you curate the project.

☒ place_guess	☒ latitude	☒ longitude	☒ positional_accuracy
☒ private_place_guess	☒ private_latitude	☒ private_longitude	☒ public_positional_accuracy
☒ geoprivacy	☒ taxon_geoprivacy	☒ coordinates_obscured	☒ positioning_method
☒ positioning_device	☒ place_town_name	☒ place_county_name	☒ place_state_name
☒ place_country_name	☒ place_admin1_name	☒ place_admin2_name	

Taxon (All | None)

☒ species_guess	☒ scientific_name	☒ common_name	☒ iconic_taxon_name
☒ taxon_id			

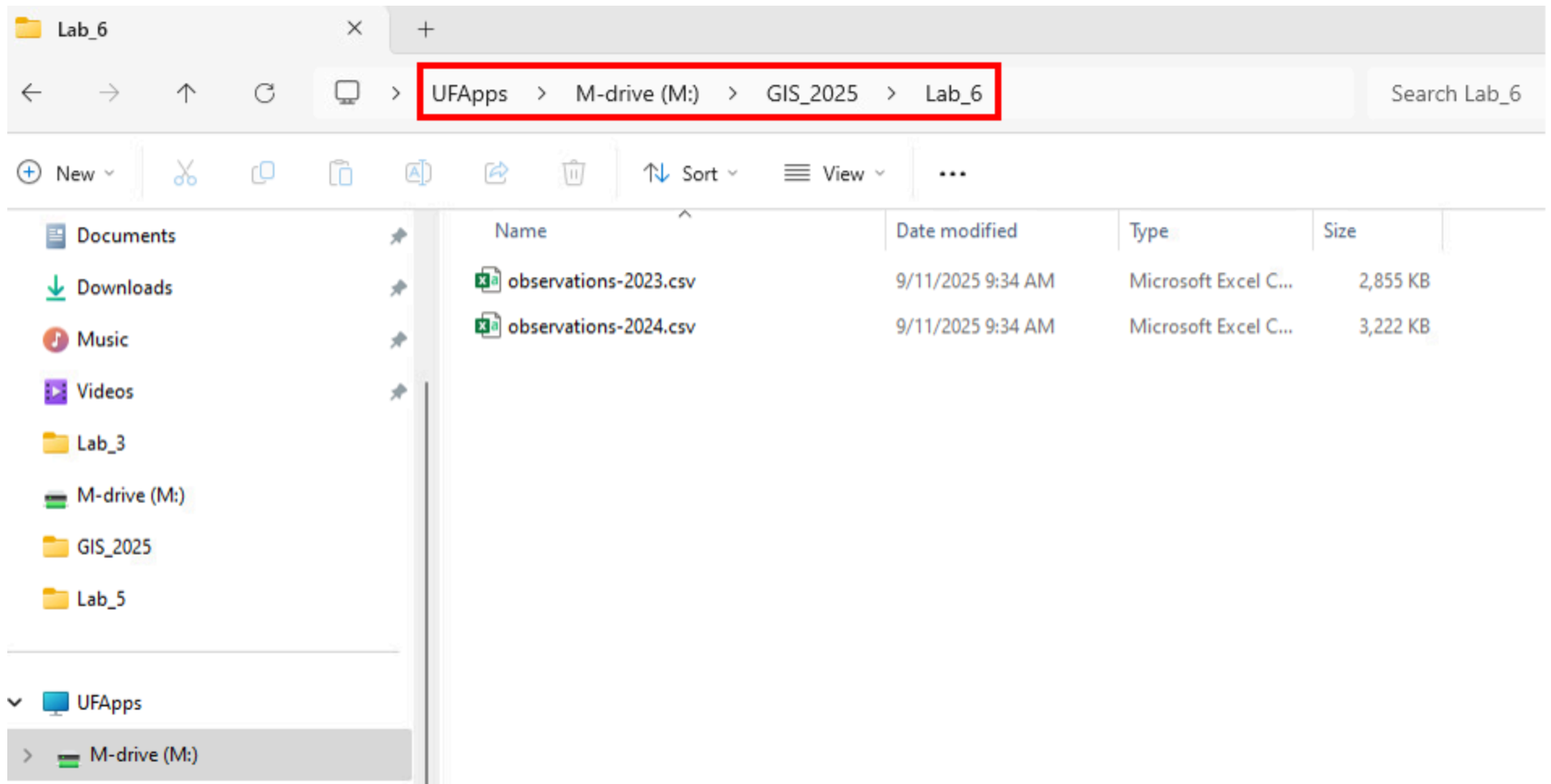
5. Click **Create Export** to download the file
6. Repeat these steps to download rattlesnake data for the date range **2024-01-01 to 2024-12-31**.

2. ArcGIS Pro

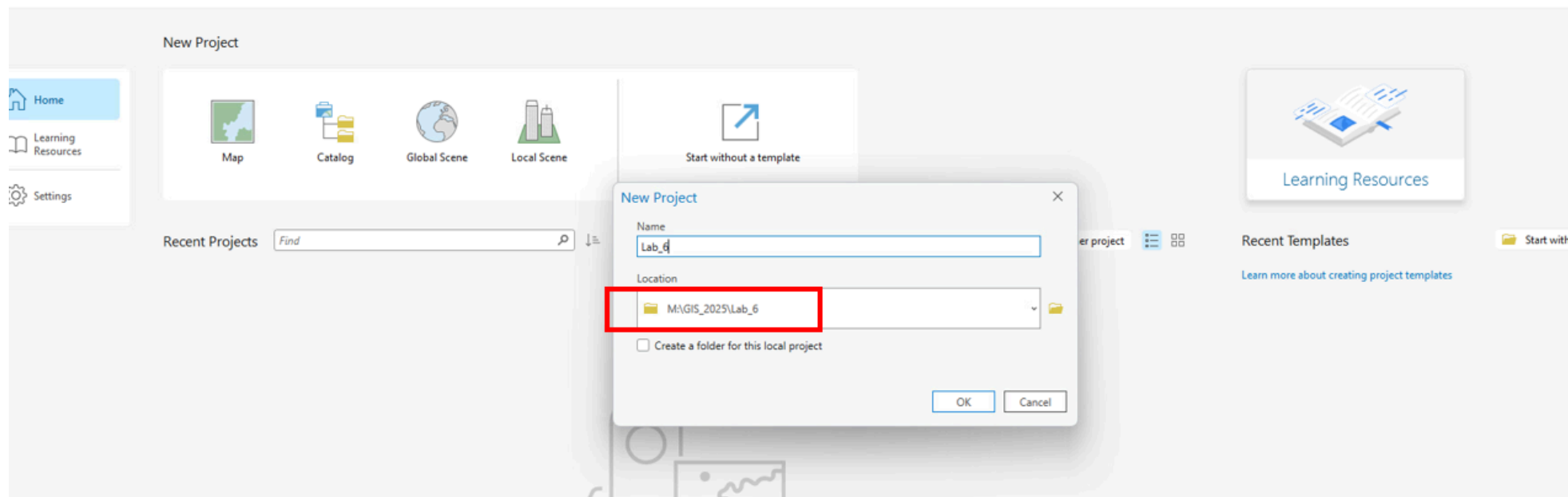
Now that we have rattlesnake distribution data for 2023 and 2024:

- We will visualize the observations in ArcGIS Pro.
- We will show point concentrations using Thiessen polygons and Kernel Density Estimation.
- We will perform a change detection analysis to highlight differences between years.

7. Copy the CSV files into your lab folder.



8. Open ArcGIS Pro and create a new project. Save the project in the same folder as your CSV files.

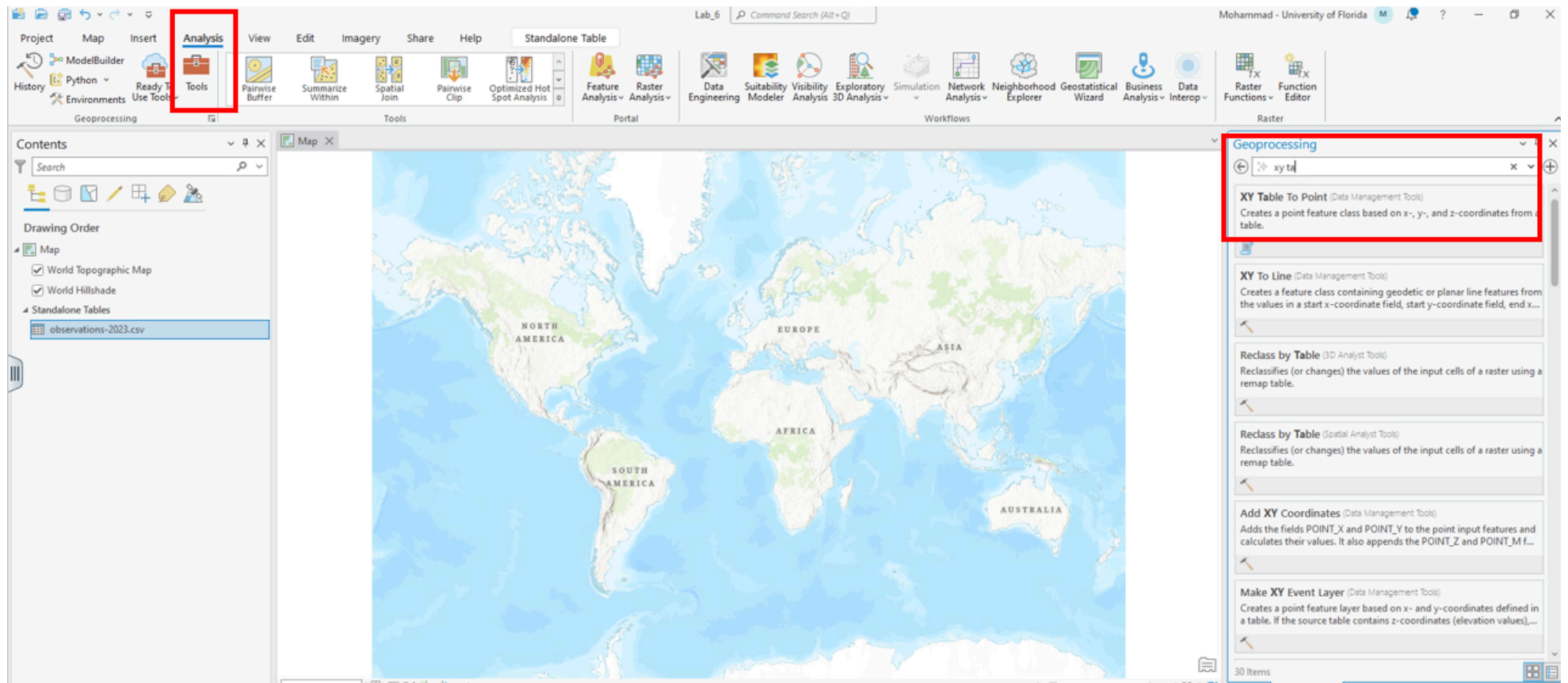


9. Add the observations-2023.csv to the map by dragging it into the Map view. You can also add the data using the Add Data > Browse option.

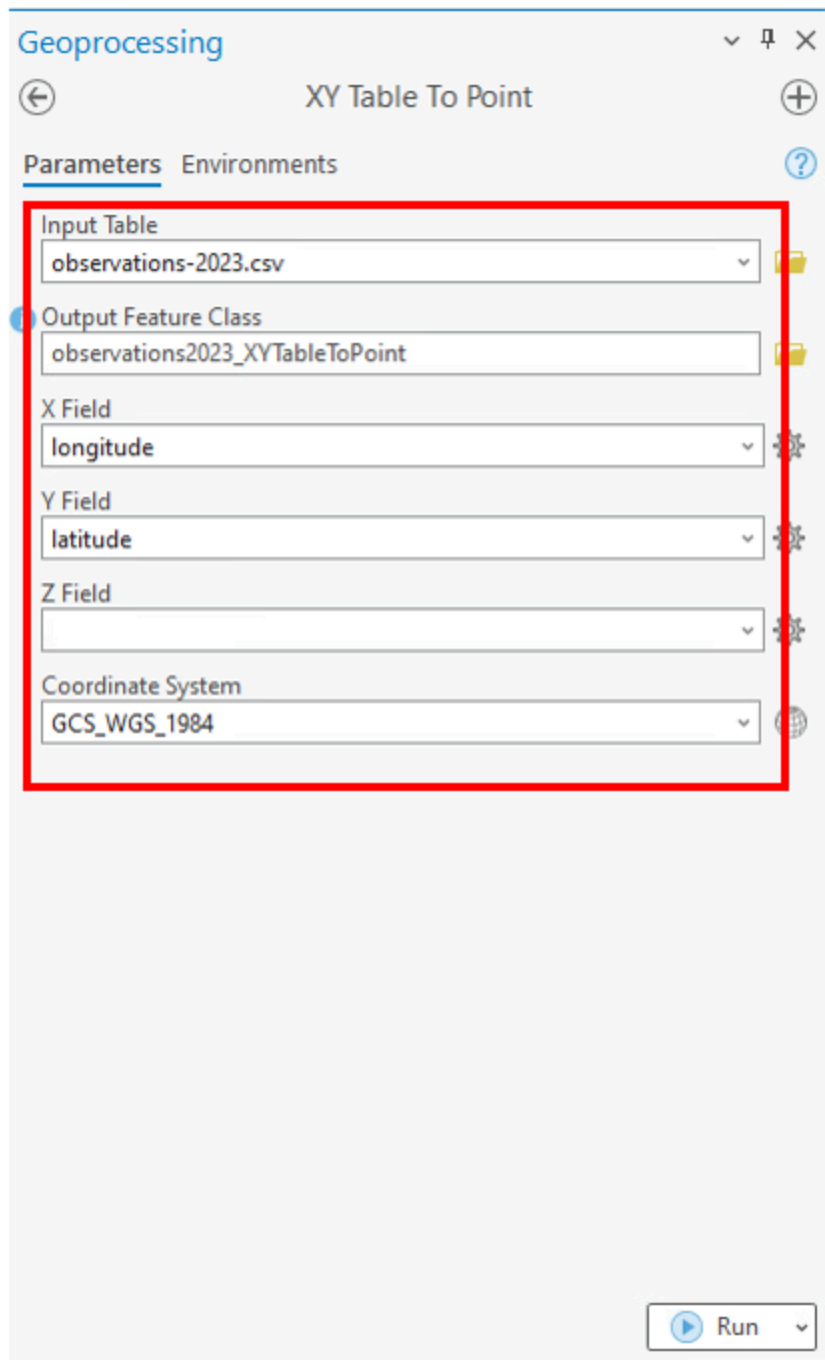


We'll use the "XY Table To Point" tool to create spatial points from the CSV file.

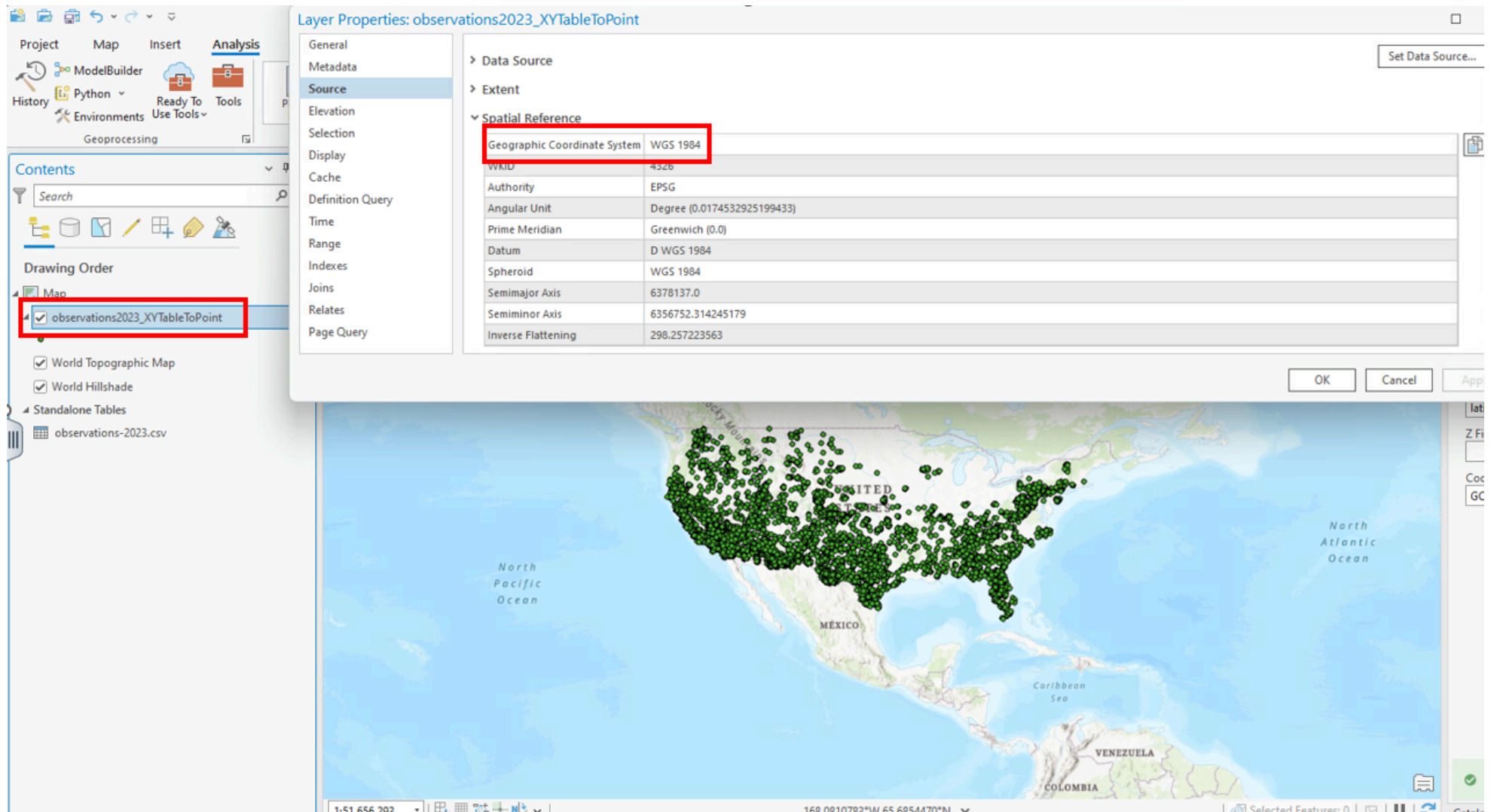
10. In ArcGIS Pro, go to Analysis > Tools, then search for "XY Table To Point" in the Geoprocessing window.



11. Select **observations-2023.csv** as the Input Table, set **longitude** as the X Field and **latitude** as the Y Field, choose **GCS_WGS_1984** for the Coordinate System, and click **Run**.

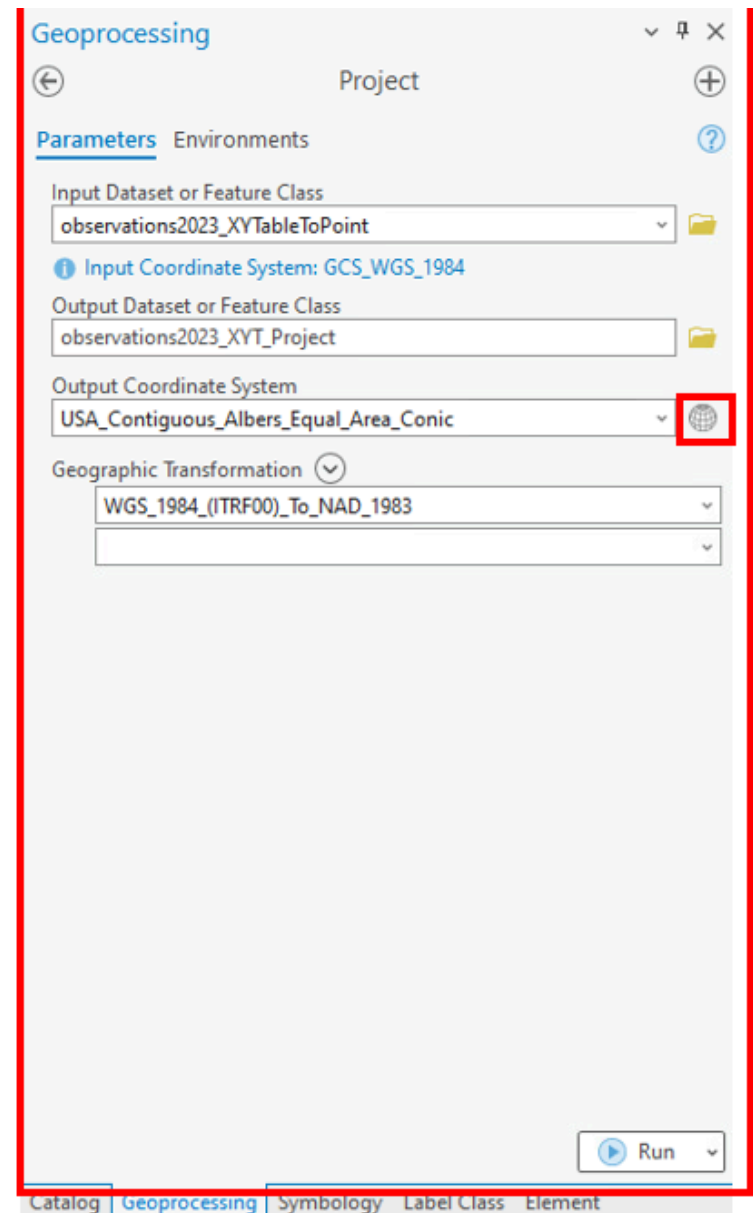
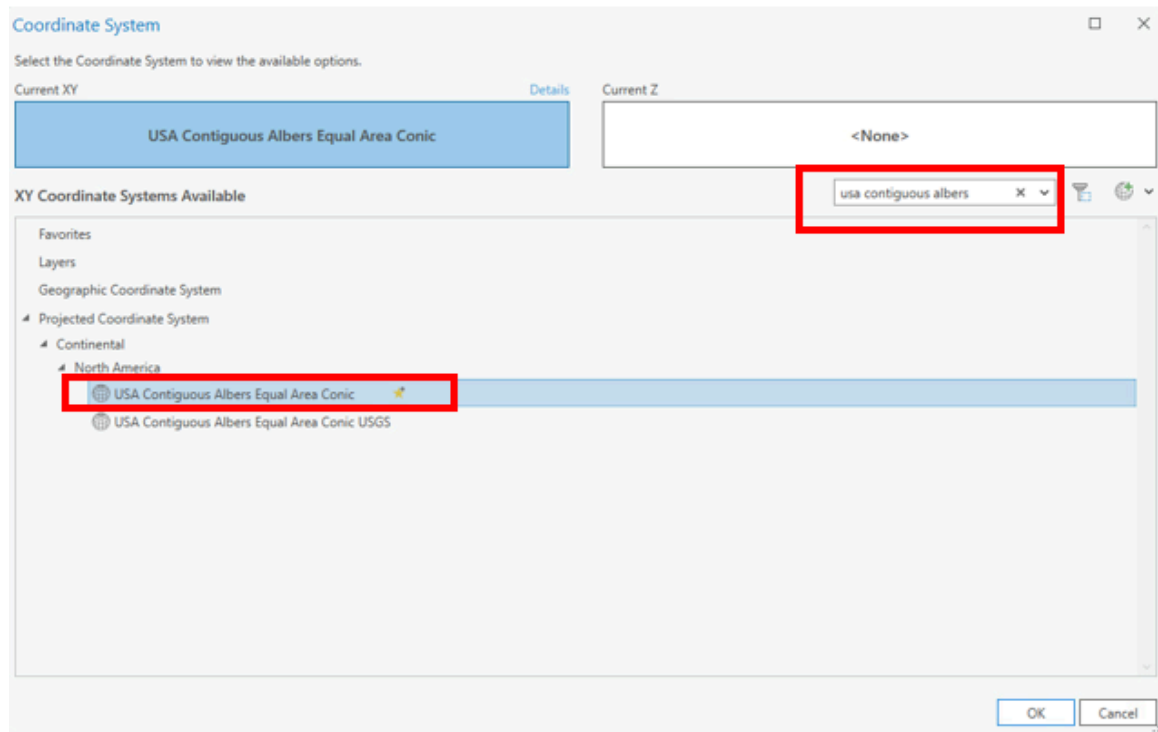


12. This will create and add **observations2023_XYTableToPoint** to the map. The new layer uses a geographic coordinate system (GCS), which you can verify by right-clicking the layer in the Contents pane, selecting Properties, and checking under Source.



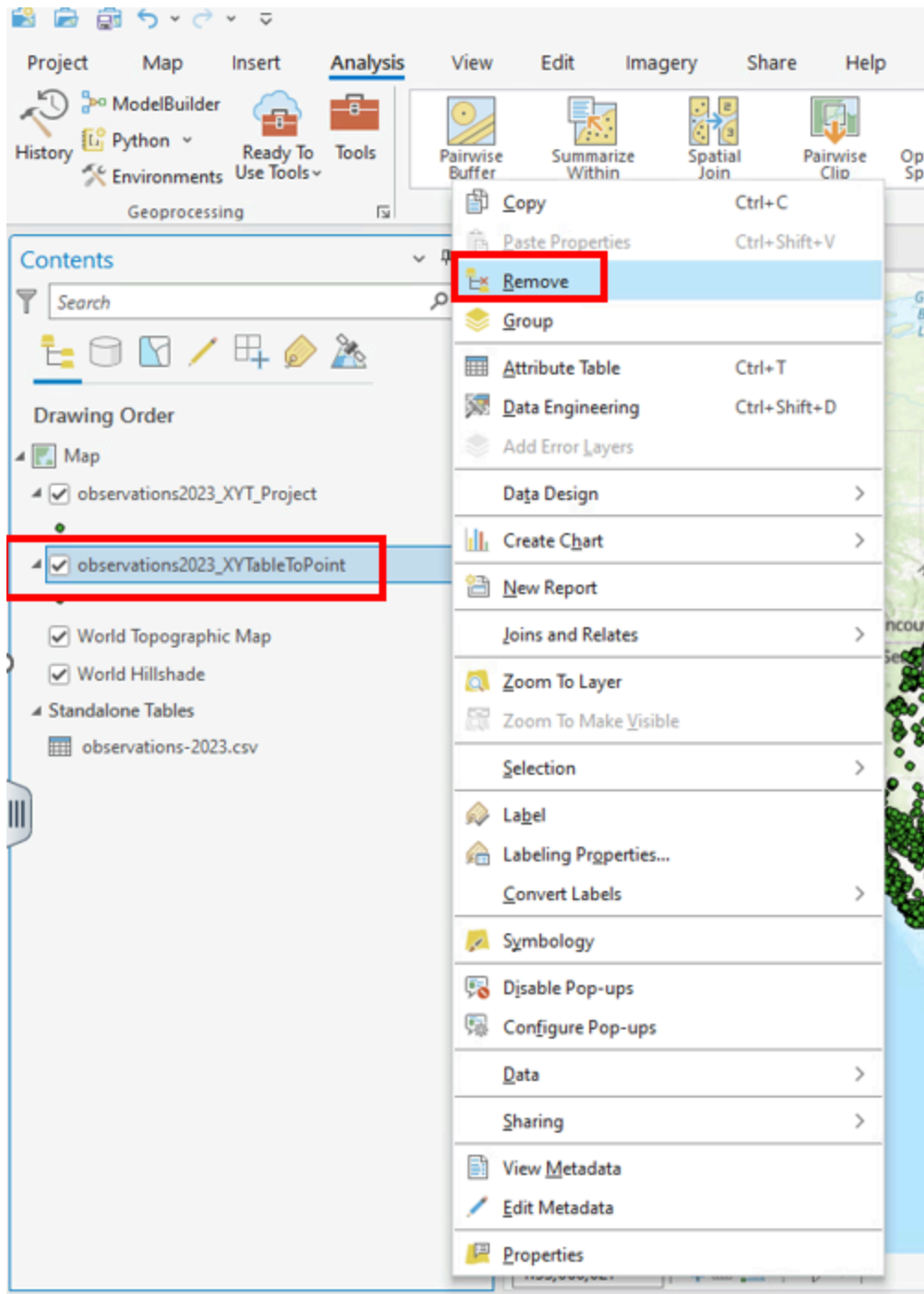
As covered in Week 3, GCS is not suitable for spatial analysis or accurate measurements. Therefore, use the Project tool to convert your data to a projected coordinate system. Here, we will use **USA Contiguous Albers Equal Area Conic**

13. Open the Project tool by going to Analysis > Tools and searching for "Project" in the Geoprocessing window. Set the Output Coordinate System to **USA Contiguous Albers Equal Area Conic**, then click **Run**.

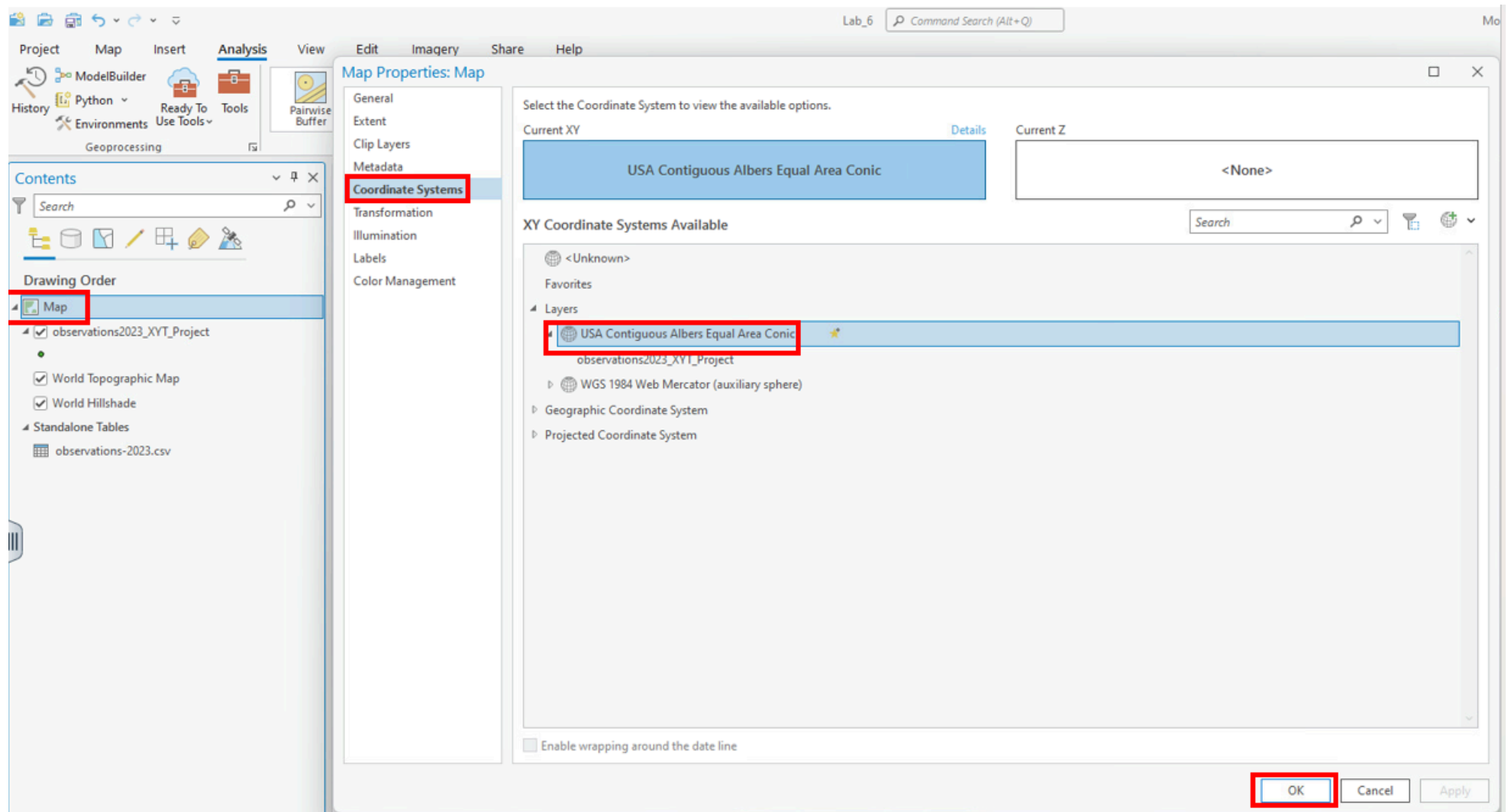


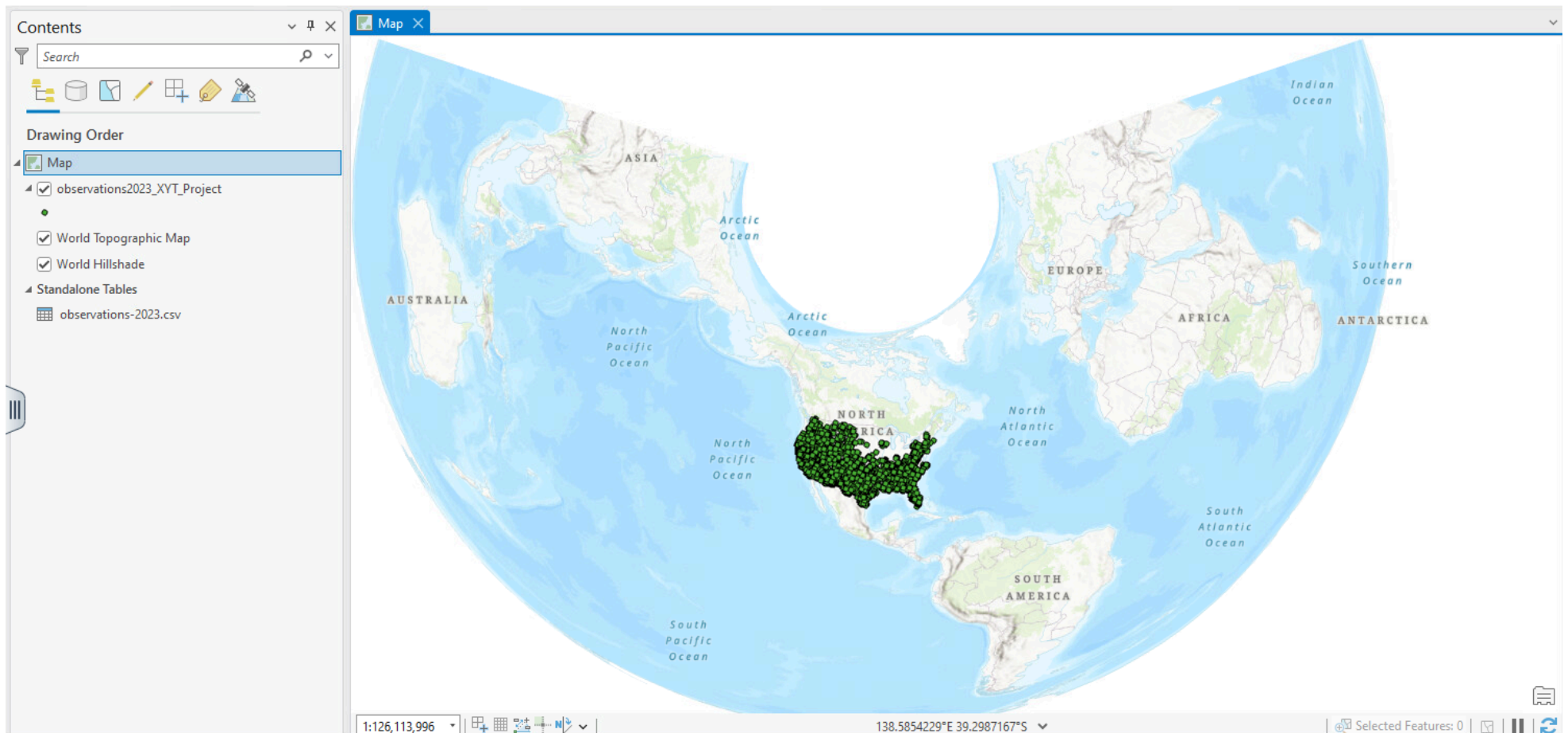
This will add a new layer, **observations2023_XYT_Project**, to the Contents pane.

14. Remove the old layer (**observations2023_XYTableToPoint**) from the Contents pane by right-clicking it and choosing **Remove**.



15. Change the projection of the basemap to match the projected coordinate system. Right-click on **Map** in the Contents pane, go to **Properties > Coordinate Systems**, select **USA Contiguous Albers Equal Area Conic**, and click **OK**.





Mapping Concentrations

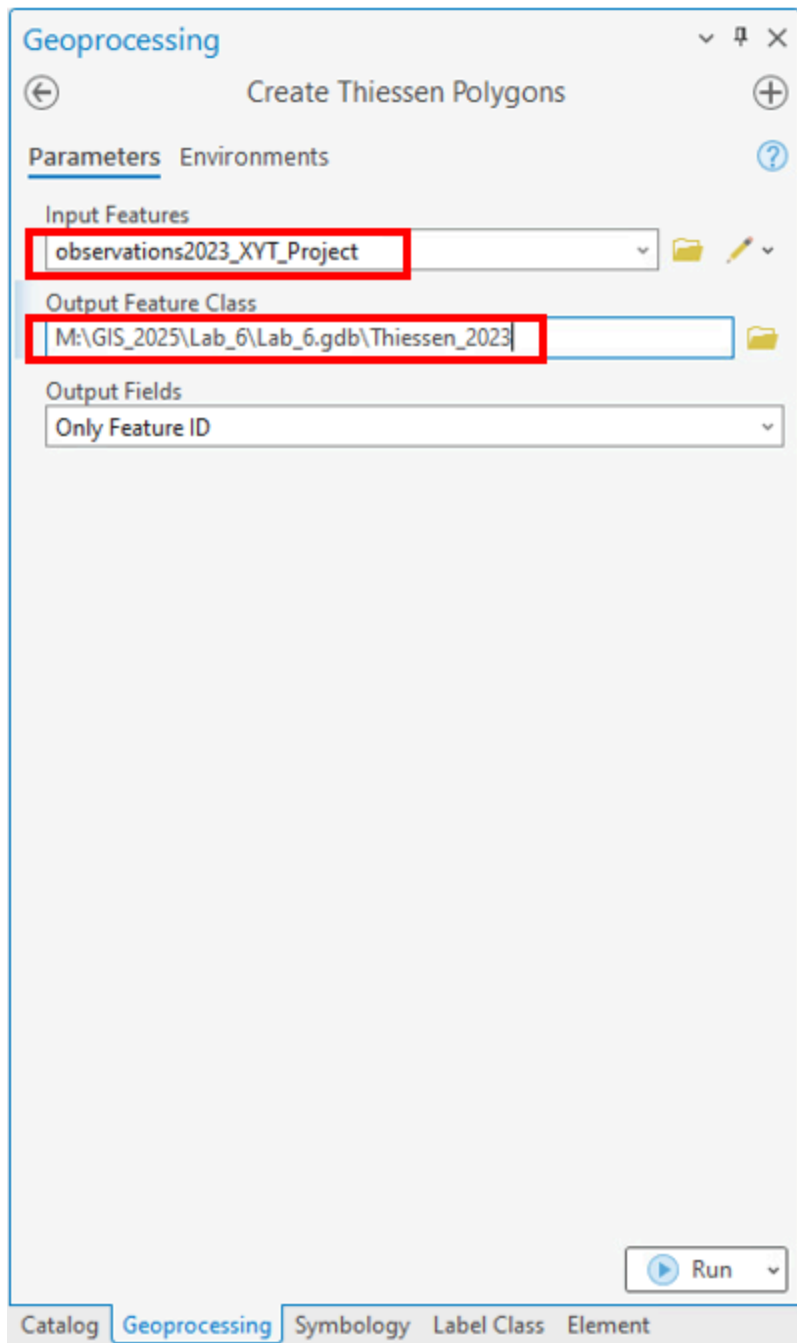
There are several ways to map concentrations of point data.

Thiessen Polygon <https://pro.arcgis.com/en/pro-app/latest/tool-reference/analysis/create-thiessen-polygons.htm>

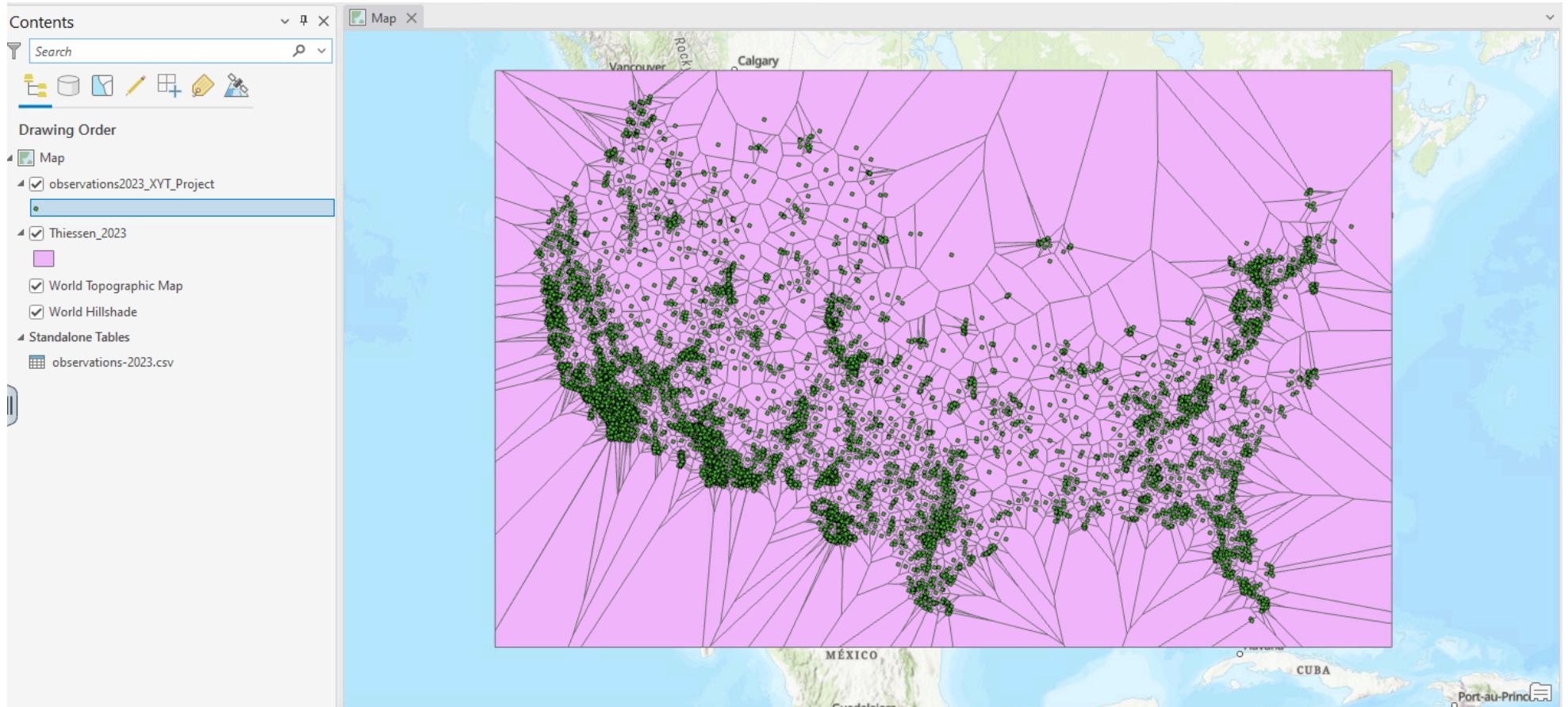
This tool divides the area covered by the input point features into polygon zones. These zones represent full areas where any location within the zone is closer to its associated input point than to any other input point. Thiessen polygons are often used to assign service areas to facilities. For example, to show the area in a city that is closest to a facility such as school or fire station.

Here we use Thiessen polygons to simply visualize point concentrations: **Smaller polygons = closer points = higher concentration**

16. Open the tool by going to Analysis > Tools and searching for **Create Thiessen Polygons**. Select your projected point data as the Input Features, set the Output Feature Class name to **Thiessen_2023** (leave the directory unchanged), and click **Run**.

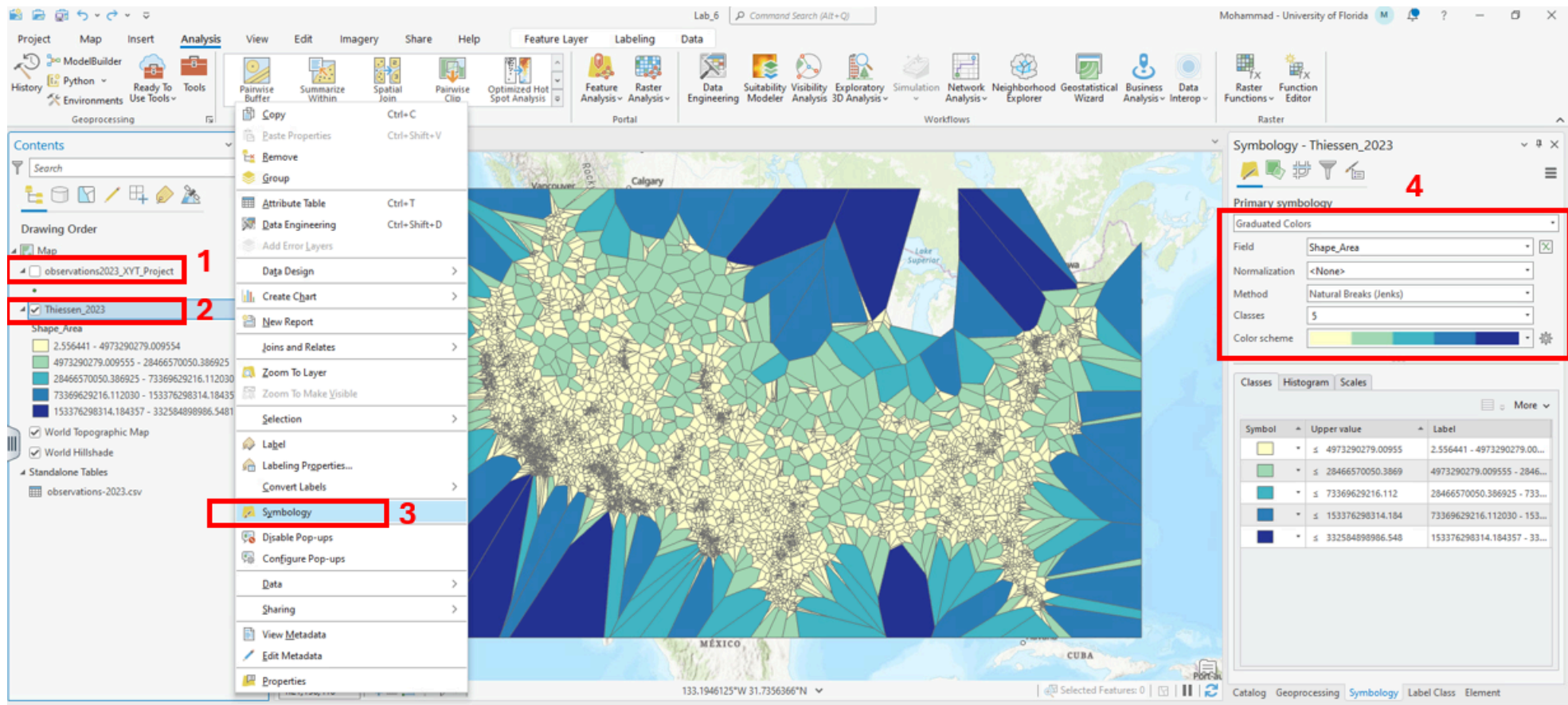


This will create a polygon around each point, representing the area where any location inside the polygon is closer to that point than to any other.



17. Unselect the point layer, then right-click on the **Thiessen_2023** layer and choose **Symbology**. Set the symbology to **Graduated Colors**, using **Shape_Area** as the field, **Natural Breaks** as the method, and set the number of classes to **5**.

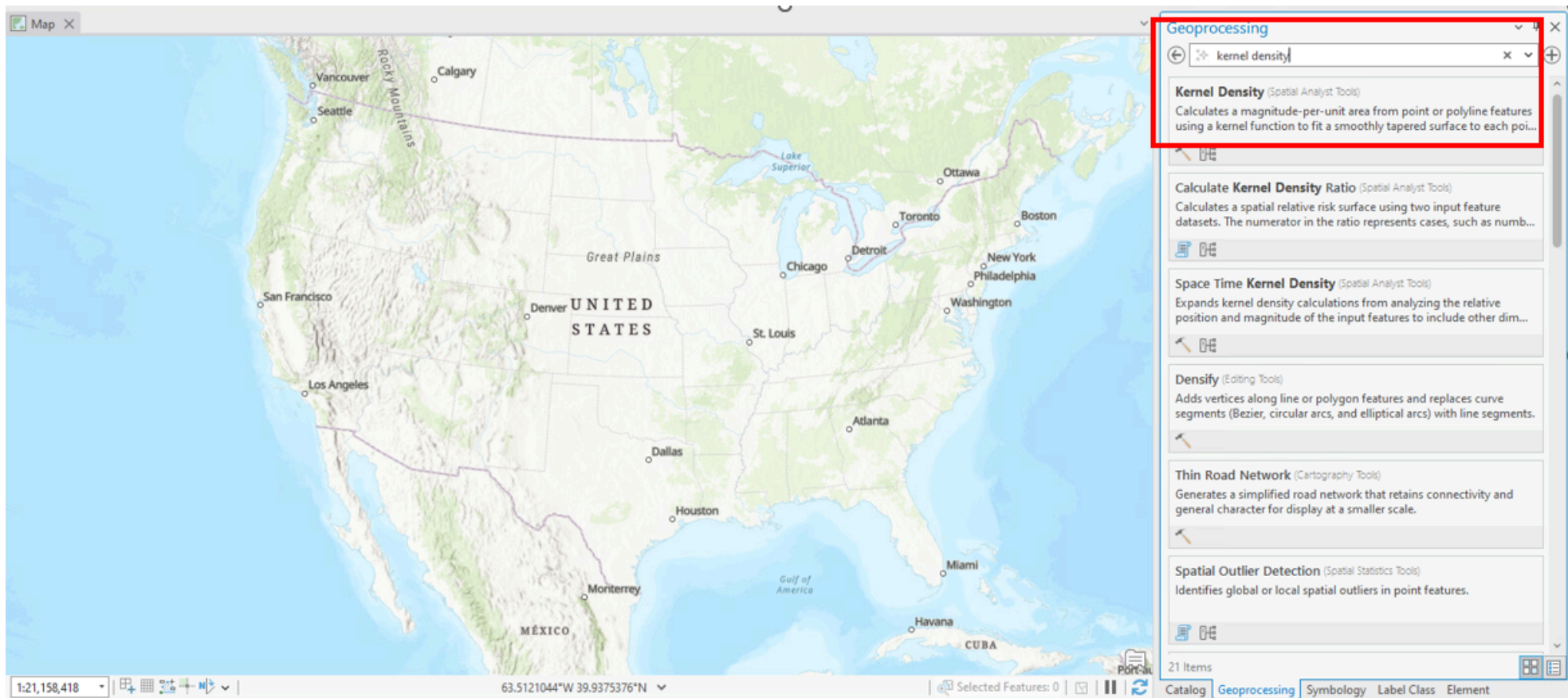
Task 1: Choose a color scheme you like and take a screenshot of the Thiessen polygons and added to the Answer sheet.



Density Estimation <https://pro.arcgis.com/en/pro-app/3.4/tool-reference/spatial-analyst/kernel-density.htm>

To map concentrations more quantitatively, we'll use Kernel Density Estimation. This tool calculates point density across the landscape by converting discrete point data into a continuous raster surface, highlighting areas of higher concentration.

18. In the Geoprocessing window, search for **Kernel Density**.



19. Select the projected 2023 point layer as the input, set the Output raster name to **KernelID_2023** (keep the default directory), leave the other options as default, and click **Run**.

Geoprocessing



Kernel Density



Parameters Environments



Input point or polyline features

observations2023_XYT_Project



Population field

NONE



Output raster

M:\GIS_2025\Lab_6\Lab_6.gdb\KernelD_2023



Output cell size

11192.7192476



Search radius

Area units

Square kilometers



Output cell values

Densities



Method

Planar



Input barrier features

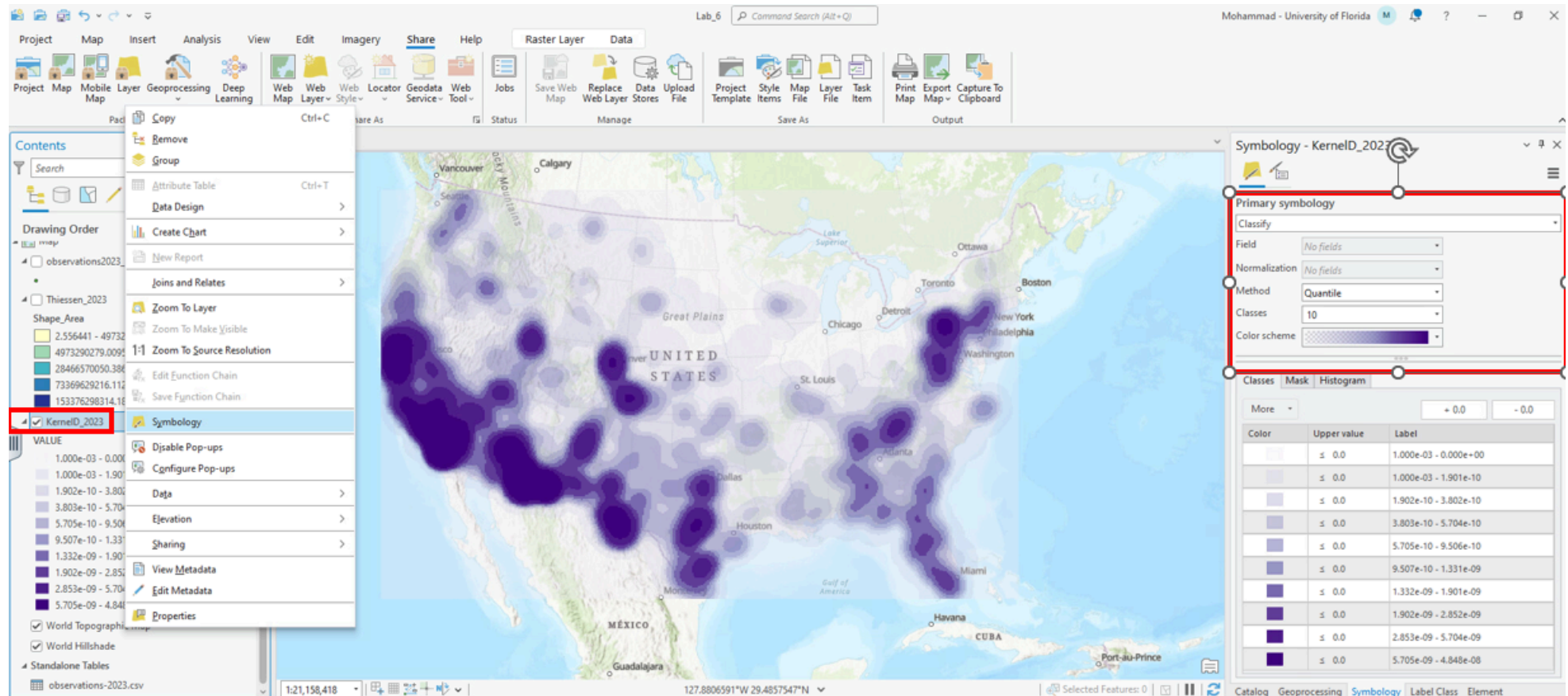


Run



20. This creates a raster layer where each pixel value represents point concentration—higher values mean greater density in that area. Right-click **KernelID_2023**, select **Symbology**, classify the pixel values using the **Quantile** method with 10 classes.

Task 2: Choose a color scheme you like and take a screenshot of the Kernel Density result and added to the Answer sheet.



21. Repeat steps 9 to 20 with **observations-2024.csv**.

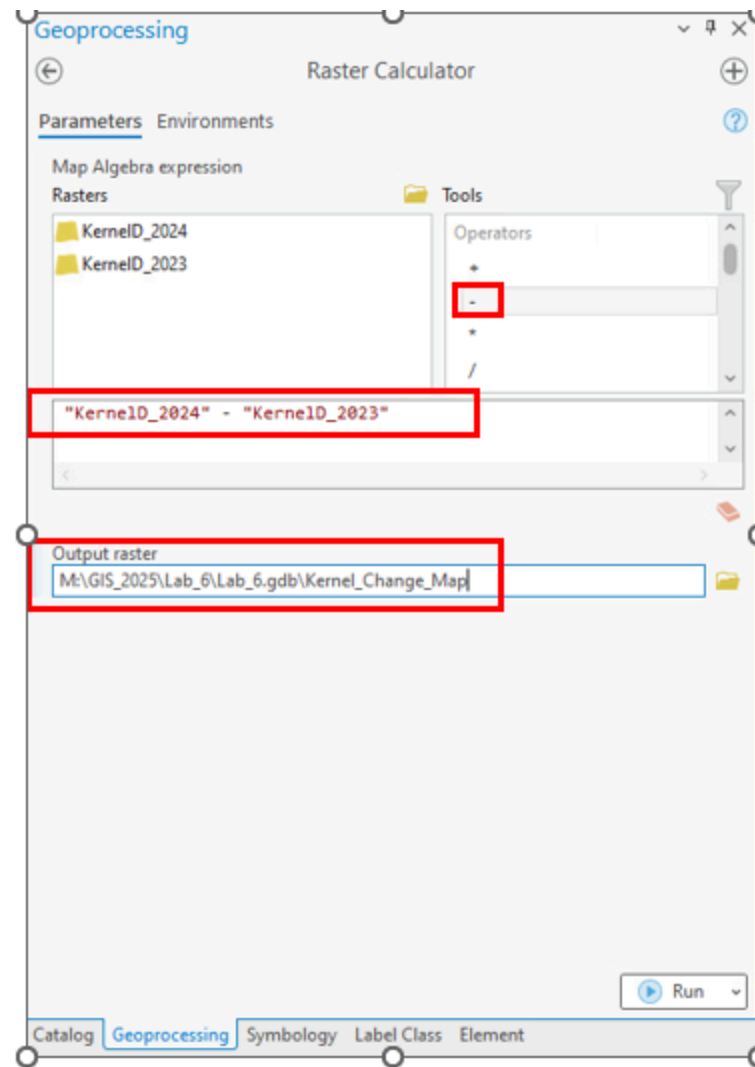
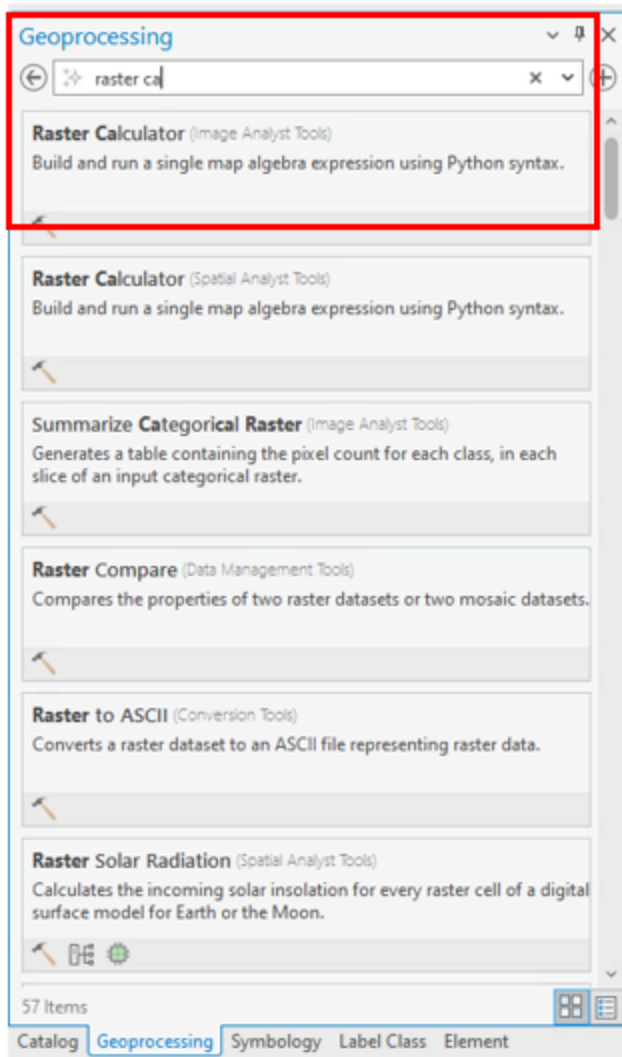
Tasks 3 and 4: Take screenshots of the Thiessen polygon and Kernel Density results of 2024 data, and add them to your Answer sheet.

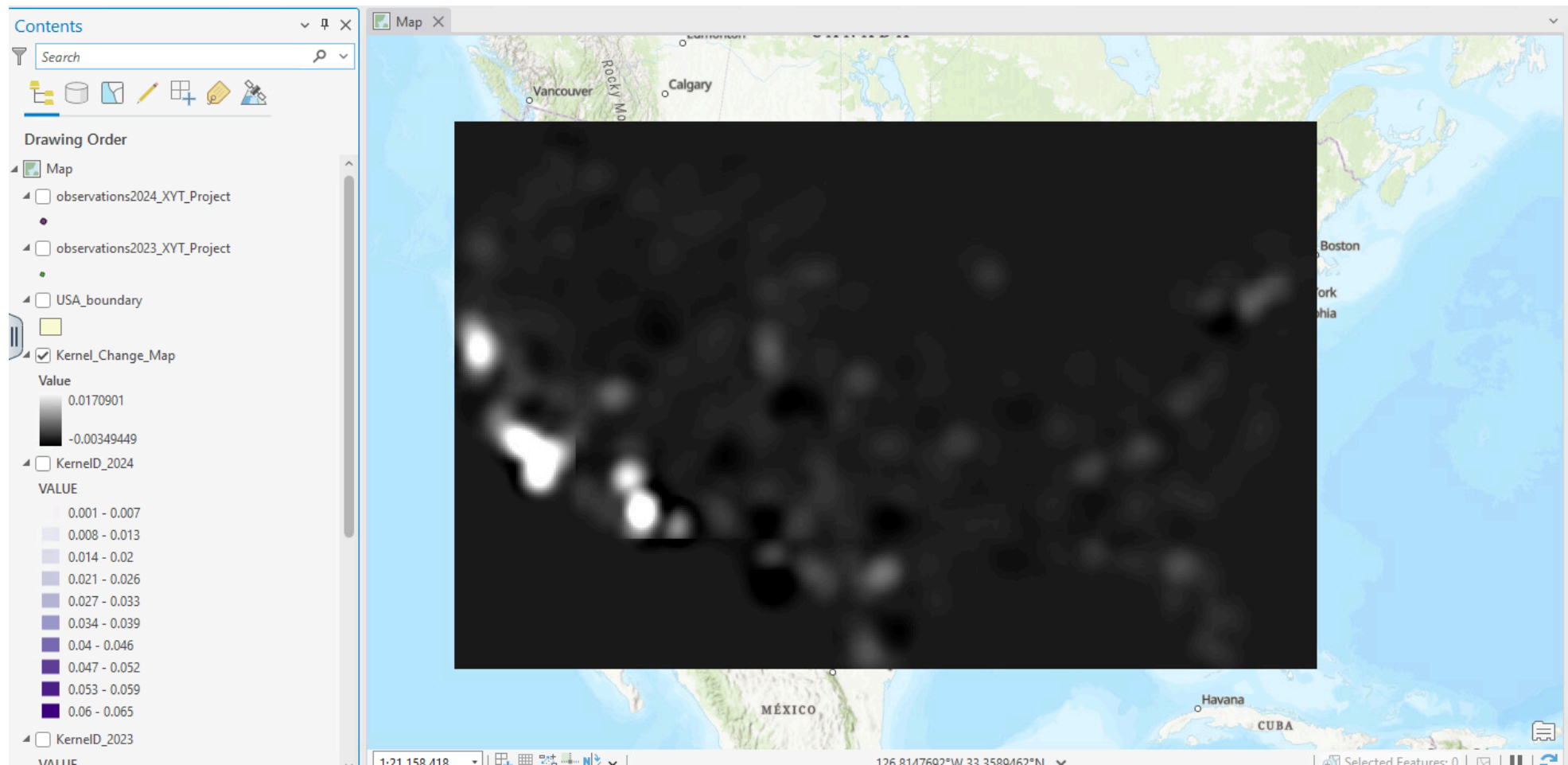
After completing the process for the 2024 data, you will have two kernel density maps: **KernelID_2023** and **KernelID_2024**.

Species distribution change detection

22. To analyze changes, we will create a new raster layer by subtracting the 2023 kernel density map from the 2024 kernel density map.

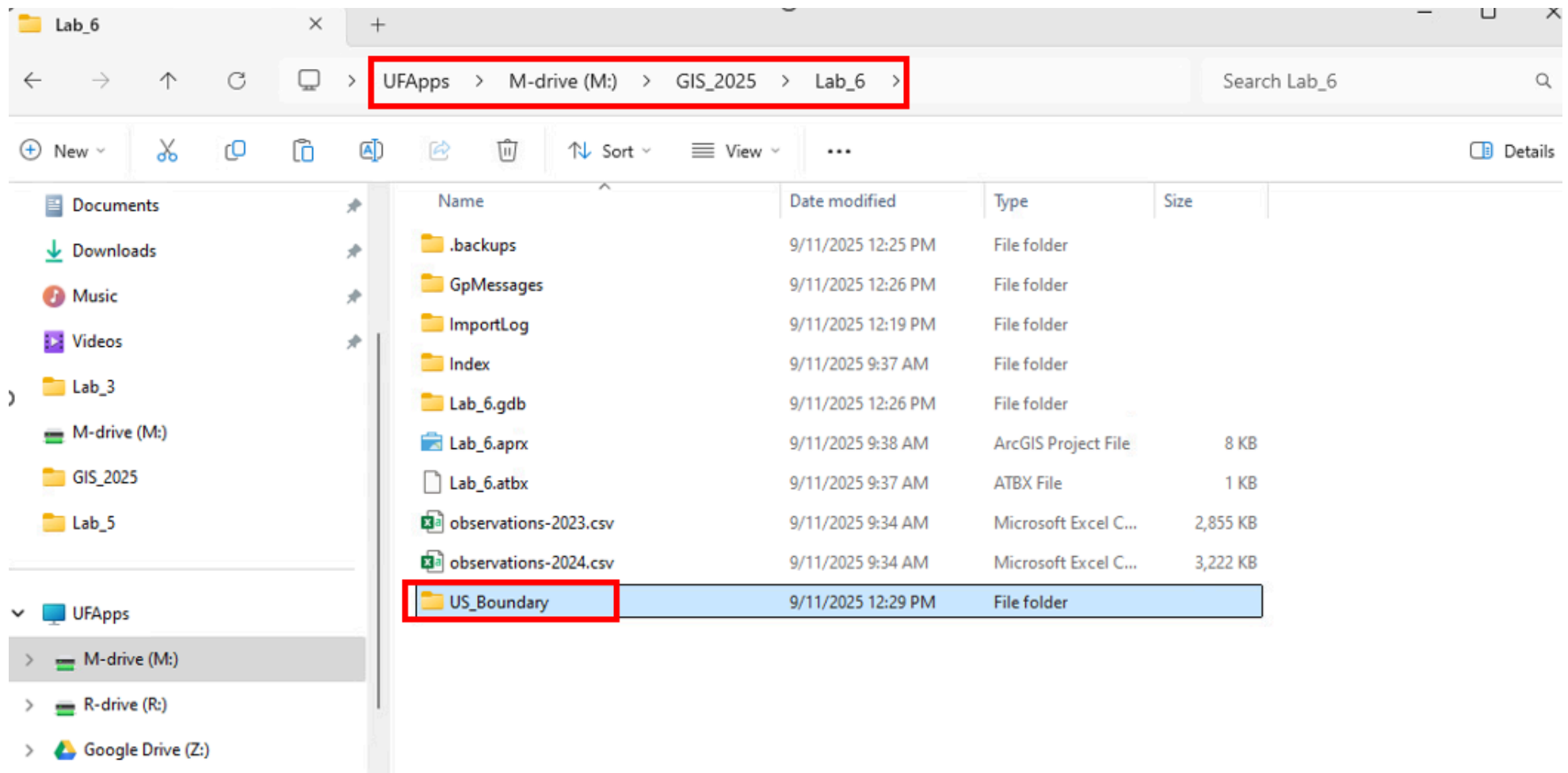
- In the Geoprocessing window, search for **Raster Calculator** (Image Analysis Tools).
- Build the expression **KernelID_2024 - KernelID_2023** by double-clicking the layer names to add them.
- Set the output raster name to **Kernel_Change_Map** (leave the directory unchanged).
- Click **Run** to generate the change detection map.



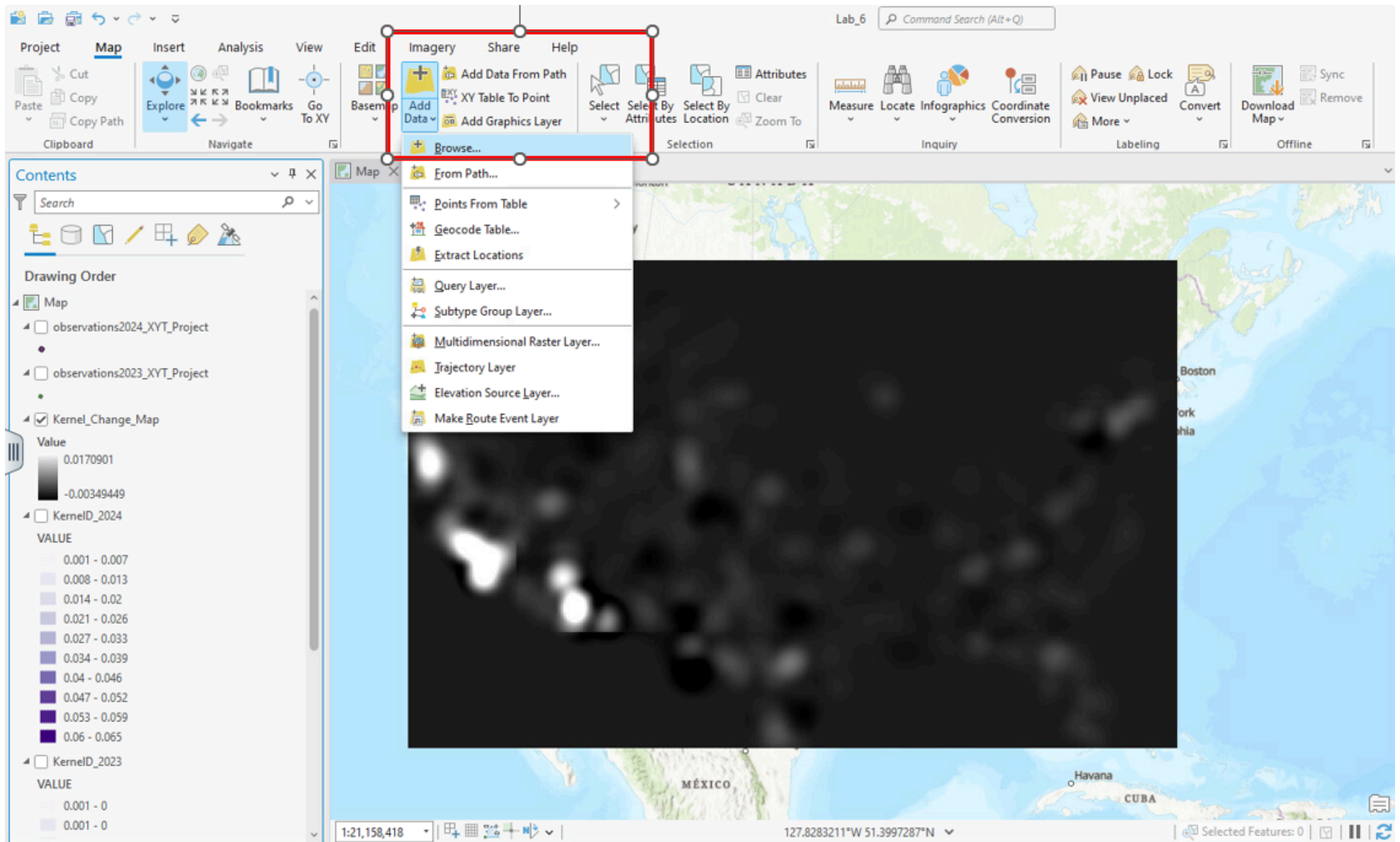


23. To clip the output raster to the USA boundary, download the **USA_boundary.zip** shapefile, unzip it, and copy the extracted folder into your lab folder.

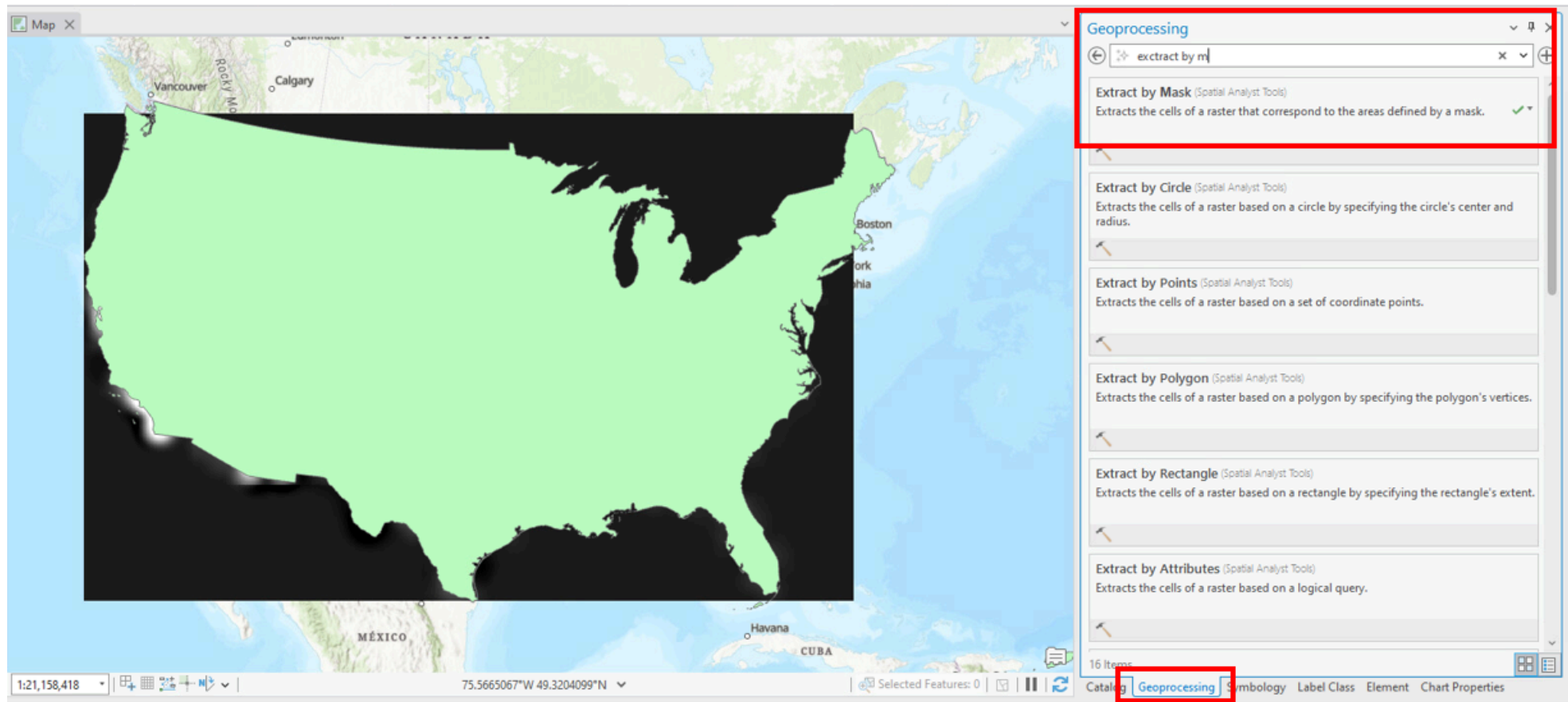
[USA_boundary.zip \(https://ufl.instructure.com/courses/541369/files/99943956?wrap=1\)](https://ufl.instructure.com/courses/541369/files/99943956?wrap=1) ↓
(https://ufl.instructure.com/courses/541369/files/99943956/download?download_frd=1)



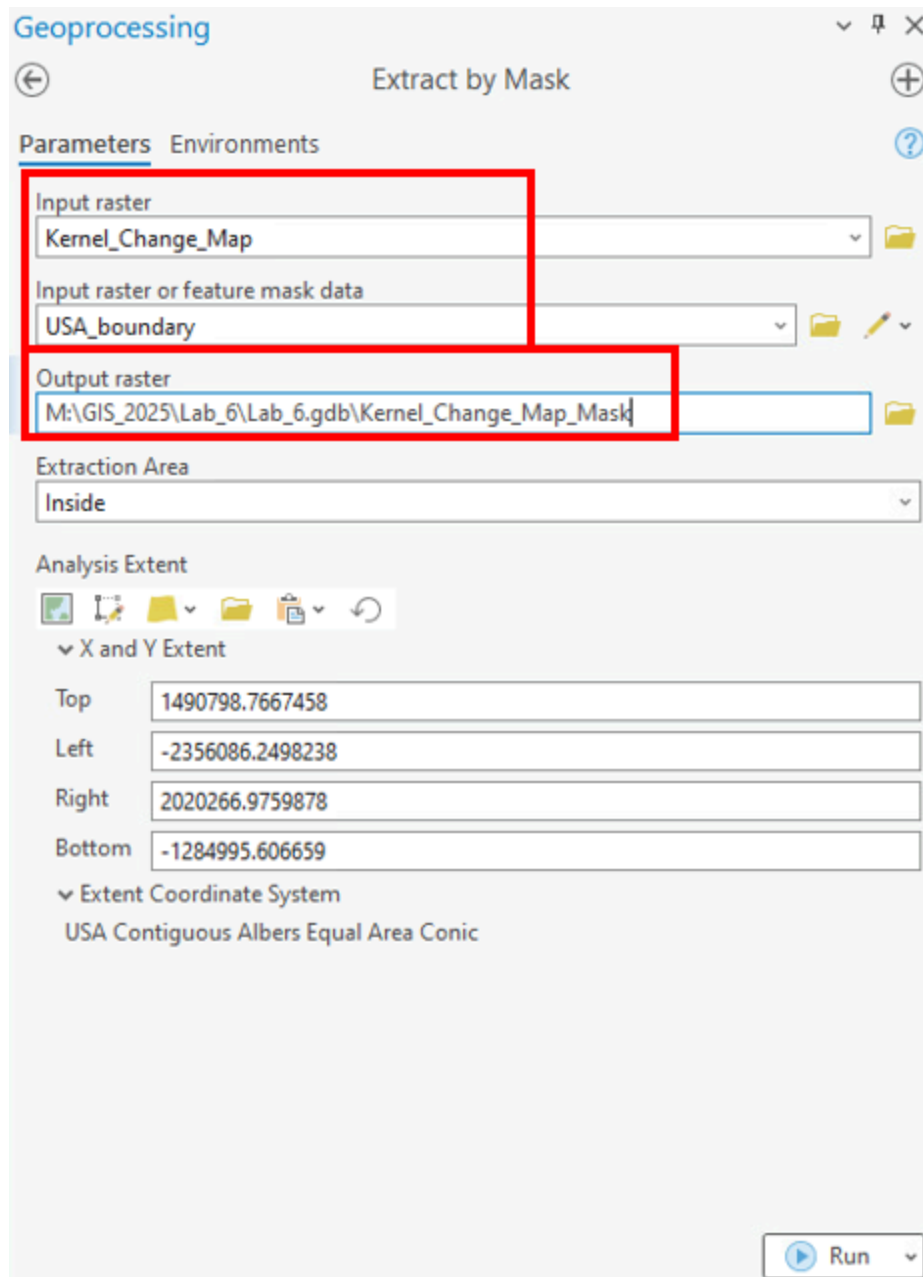
24. Add the USA_boundary.shp to the map by dragging it from the Catalog pane into the Map view. You can also add the data from the Map tab, **Add Data > Browse**, then select **USA_boundary.shp** and add it to your map (If you don't see **USA_boundary.shp**, right-click inside the directory window and choose **Refresh**.).



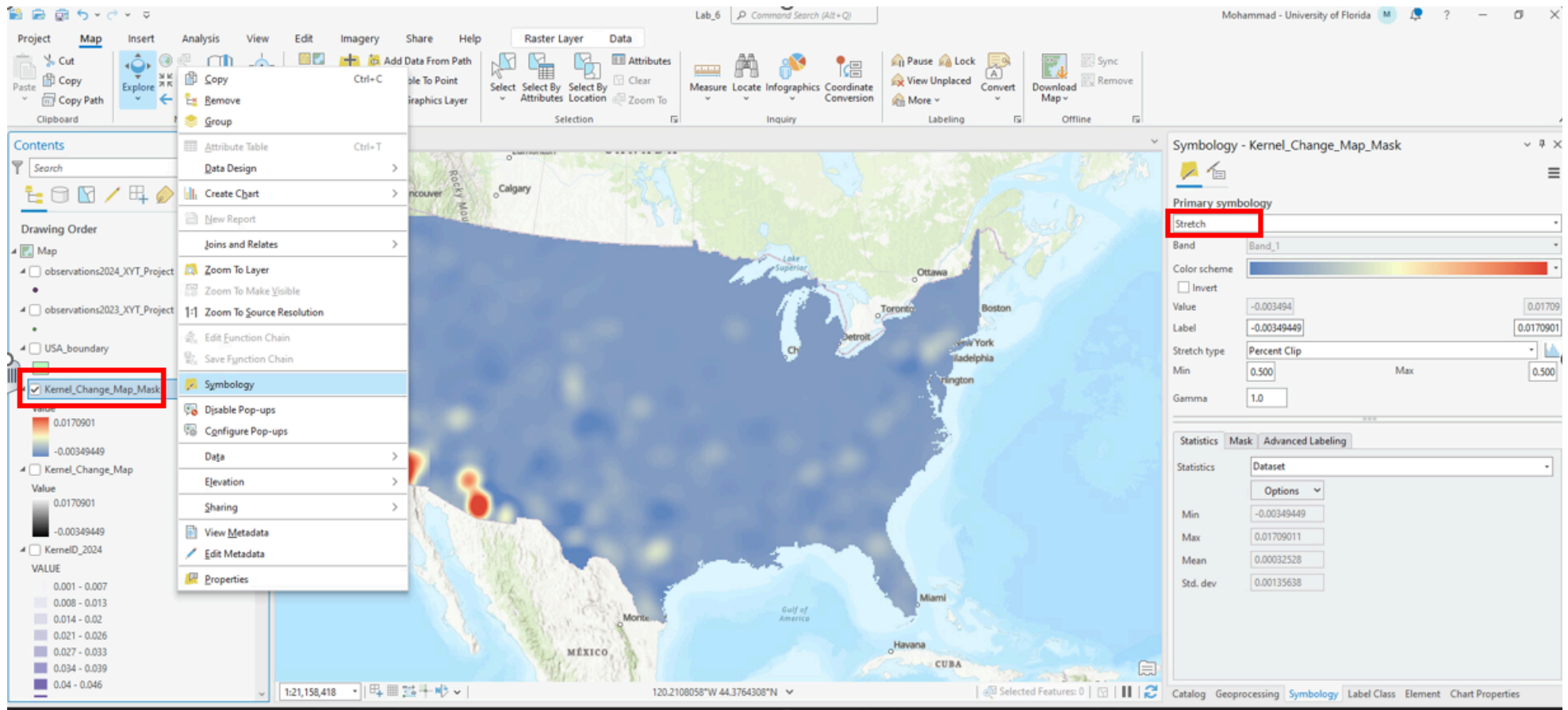
25. In the Geoprocessing window, search for **Extract by Mask** tool.



26. Select **Kernel_Change_Map** as the input raster, **USA_boundary** as the mask layer, name the output raster **Kernel_Change_Map_Mask**, and click **Run**.



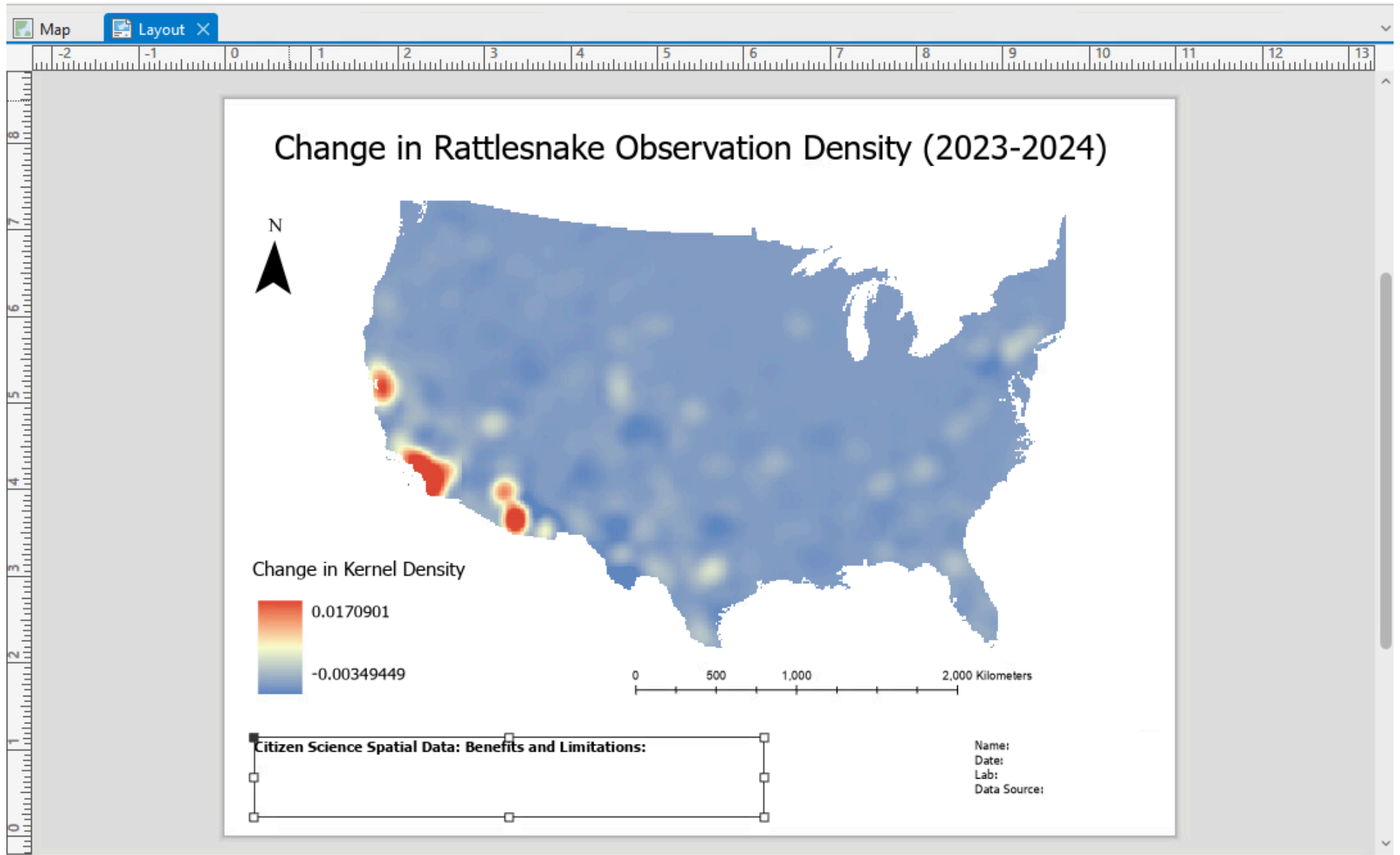
27. Unselect all other layers in the Contents pane. Right-click the created masked layer, go to **Symbology**, select **Stretch**, and choose a color scheme you like.



In this layer, positive values indicate an increase in point concentration from 2023 to 2024, negative values indicate a decrease, and values near zero represent areas with little or no change in density.

28. Make a map of the distribution change:

- From the Insert tab, create a new layout and add your map frame.
- Include a title, legend, scale bar, north arrow, and your identification (name, date, etc.) on the map.
- Add a brief paragraph to the layout discussing the pros and cons of citizen science–derived spatial data.



Your submission should include:

- Screenshots of Thiessen polygons for 2023 and 2024 (3 pts)

- Screenshots of kernel density maps for 2023 and 2024 (3 pts)
- Final map layout showing distribution change (9 pts)

Have Fun Mapping Species Distribution!

Points 15

Submitting a file upload

Due	For	Available from	Until
Sep 21	Everyone	Sep 16 at 12am	Sep 26 at 11:59pm

+ Rubric